Sachintha Dilshan

Software Engineering Intern

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Professional Summary

Third-year Information Technology undergraduate at the University of Moratuwa with hands-on experience building distributed backend systems, full-stack web applications, and AI-powered solutions. Skilled in designing REST APIs, authentication and access-control systems, concurrent job processing, database-driven applications, and retrieval-augmented AI systems. Passionate about applying software engineering principles to build reliable, maintainable, and well-structured solutions and seeking a Software Engineering Internship to contribute technical skills in an industry environment.

Education

University of Moratuwa – B.Sc. (Hons) in Information Technology *2024 – 2028 (Expected)*

CGPA: 3.59 / 4.00

Rahula College, Matara – G.C.E. Advanced Level Examination (Physical Science Stream) *2022*

Z-Score: 2.065

Technical Skills

Languages – Java, Python, TypeScript, JavaScript, SQL, C

Web & Backend – Spring Boot, FastAPI, Express.js,Node.js, Next.js, React, REST APIs, JWT Authentication, Role-Based Access Control, Prisma, SQLAlchemy

Databases – PostgreSQL, SQLite, ChromaDB, IndexedDB

AI & Machine Learning – RAG Pipelines, LLM Integration, Sentence Transformers, XGBoost, LightGBM

DevOps & Tools – Docker, Git, GitHub, CI/CD, Flyway, JUnit, Prometheus, Micrometer

Projects

CoreHR – Human Resource Management System *[Group]*

FastAPI, Next.js, PostgreSQL, SQLAlchemy, Alembic GitHub

- Developed the document management module for a full-stack HR platform covering recruitment, employee records, leave, and document workflows.
- Implemented approval workflows with separation-of-duties enforcement and automated PDF document generation.
- Built an IMAP-based background service for processing inbound HR requests while preventing duplicate records.

Distributed Job Queue – Reliable Background Job Processing *[Individual]*

Java, Spring Boot, PostgreSQL, Docker, Flyway, JUnit, GitHub Actions, Prometheus GitHub

- Developed and published a reusable Spring Boot library (jobqueue-core) to GitHub Packages, providing concurrent background job processing across multiple workers with a pluggable handler interface.
- Implemented worker leases, heartbeats, exponential backoff retries, automatic crash recovery, and dead-letter handling — verified with JUnit integration tests, GitHub Actions CI, and a live Prometheus metrics endpoint.

AAAC – Access-Aware Admission Control *[Group]*

Python, FastAPI, Redis, LightGBM, Docker GitHub

- Developed a fairness-aware virtual waiting room that adapts admission windows and page payloads to client link quality, reducing timeouts for users on slow connections.
- Implemented LightGBM-based link estimation, adaptive payload delivery, client SDK, and live dashboard components, with Docker-based network emulation and automated evaluation.

Rahula Dancing Society Platform – Learning Portal *[Individual]*

Next.js, React, Tailwind CSS, Express.js,Node.js, Prisma, PostgreSQL, JWT GitHub

- Developed a full-stack learning portal for managing students, marks, quizzes, and study resources.
- Implemented JWT authentication with Admin, Teacher, and Student role-based access control and secure data isolation.
- Built server-side quiz evaluation and controlled exam access using a structured Express backend.

Mechanical Engineering Material Selection Assistant *[Individual]*

Python, ChromaDB, Sentence Transformers, SQLite, Streamlit GitHub

- Developed a RAG-based assistant that recommends engineering materials from natural-language operating conditions.
- Designed semantic retrieval with verified material properties stored in SQLite as the source of truth.
- Integrated multi-provider LLM support with response caching and provider failover.

Sentrio – AI Browser Extension for Social Engineering Detection *[Group]*

JavaScript, Chrome Manifest V3, Groq, Llama 3.3, IndexedDB GitHub

- Developed the AI reasoning layer for detecting social engineering and phishing attacks targeting Sri Lankan banking users.
- Integrated multi-provider LLM support with evidence grounding from trusted external sources.
- Achieved **2nd Runner-Up** at Aurora 2026, University of Sri Jayewardenepura.

Certifications

AWS Cloud Practitioner Essentials *Amazon Web Services*

Introduction to DevOps and Site Reliability Engineering (LFS162) *The Linux Foundation*

Achievements

1st Runner-Up – COMFIX 2026, University of Moratuwa *Team Vortex*

2nd Runner-Up – Aurora 2026, University of Sri Jayewardenepura *Team Mora Vortex*

1st Place, Round 1 – OctWave 3.0 Kaggle Hackathon *Team Mora Vortex*

Finalist – Codesplash AI Agent Hackathon 2026, University of Kelaniya *Team Mora Vortex*

Leadership & Volunteering

IT & Media Coordinator – University of Moratuwa Dancing Society *2026 – Present*

Organizing Committee Member – FIT Future Career Fair *2025 – 2026*

Managed digital content for society activities and contributed to organizing university career events.

References

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